

WOLF

ABLATION
IV BB/CCB
ADENOSINE
VALSALVA

WIDE COMPLEX
= VT UNTIL PROVEN

REGULAR

IRREGULAR

DEFIB READY

ALWAYS AVAILABLE

PREEXCITED AF

>250 bpm

VF → SUDDEN DEATH

25% OF APs

BYSTANDER

DO NOT USE

AV BLOCK
→ ↑ AP → VF

BETA BUGLE

AF 15%

CALCI-YUM ICE CREAM

IV AMIODARONE

ALWAYS AVAILABLE

IBUTILIDE

PROCAINAMIDE

1. Pre-excited AF

WHAT YOU SEE

PREEXCITED AF banner, fireballs over castle

WHAT IT ENCODES

Pre-excited AF is atrial fibrillation in a WPW patient where the chaotic atrial impulses conduct DOWN the accessory pathway, bypassing the rate-limiting AV node. Because a manifest accessory pathway conducts rapidly and non-decrementally, the ventricular response becomes extremely fast and irregular, and the QRS is wide and pre-excited (varying degrees of fusion). This is a true emergency because it can degenerate into ventricular fibrillation.

BOARD TRAP

WRONG: 'Pre-excited AF is just rapid AF and should be rate-controlled like any other AF.' The discriminator is the accessory pathway — standard AF rate control with AV-nodal blockers removes the only competing route, funnels conduction down the fast pathway, and can precipitate VF; pre-excited AF is managed with procainamide/ibutilide or cardioversion, not nodal blockade.

2. >250 bpm

WHAT YOU SEE

>250 bpm sign

WHAT IT ENCODES

When AF conducts antegrade over a fast accessory pathway, ventricular rates can exceed 250–300 bpm because the pathway lacks the AV node's protective decremental slowing. A shortest pre-excited RR interval <250 ms during AF identifies a high-risk pathway prone to VF. The faster the conduction, the higher the SCD risk.

BOARD TRAP

WRONG: 'A ventricular rate over 250 bpm with an irregular wide-complex rhythm is polymorphic VT requiring amiodarone.' The discriminator is irregularity with varying QRS widths in a WPW patient — that pattern is pre-excited AF; treat with procainamide/ibutilide or cardioversion and AVOID AV-nodal blockers, rather than defaulting to amiodarone (which can also accelerate the rate).

3. Wide + irregular

WHAT YOU SEE

WIDE COMPLEX knight; REGULAR vs IRREGULAR

WHAT IT ENCODES

The hallmark ECG of pre-excited AF is an IRREGULARLY irregular WIDE-complex tachycardia with QRS complexes of VARYING width/morphology (changing degrees of pre-excitation beat to beat). This triad — wide, irregular, varying morphology, very fast — is essentially diagnostic and distinguishes it from monomorphic VT (wide but regular and uniform).

BOARD TRAP

WRONG: 'A wide-complex tachycardia that is irregular must be AF with bundle-branch-block aberrancy.' The discriminator is the VARYING QRS morphology and extreme rate — fixed-aberrancy AF has uniform wide QRS at a more modest rate, whereas pre-excited AF shows beat-to-beat changing QRS width because of variable accessory-pathway pre-excitation.

4. VT until proven

WHAT YOU SEE

= VT UNTIL PROVEN

WHAT IT ENCODES

Any wide-complex tachycardia is VT until proven otherwise — VT remains the most common and most dangerous cause. Within that rule, an IRREGULAR, very-fast wide-complex tachycardia with varying QRS widths points specifically to pre-excited AF. The safe acute treatment of pre-excited AF (procainamide or cardioversion) overlaps with the safe treatment of undifferentiated VT, so erring toward this pathway is also the safe default.

BOARD TRAP

WRONG: 'Because we cannot be 100% sure it is VT versus pre-excited AF, give adenosine to clarify the diagnosis.' The discriminator is that BOTH possibilities make adenosine/AV-nodal blockade dangerous — adenosine can accelerate pre-excited AF into VF and won't fix VT; use procainamide or synchronized cardioversion instead of a diagnostic nodal-blocking trial.

5. DO NOT USE

WHAT YOU SEE

Big red DO NOT USE banner

WHAT IT ENCODES

AV-nodal blocking drugs are CONTRAINDICATED in pre-excited AF. By blocking the node, they remove the competing pathway and divert all rapid atrial impulses down the accessory pathway, accelerating the ventricular rate and risking degeneration to VF. The forbidden list is captured by the mnemonic 'ABCD-A': Adenosine, Beta-blockers, Calcium-channel blockers, Digoxin, (and IV) Amiodarone. Use procainamide or ibutilide, or cardiovert.

BOARD TRAP

WRONG: 'Diltiazem or metoprolol is appropriate first-line rate control for this rapid AF.' The discriminator is pre-excitation — in pre-excited AF these AV-nodal blockers paradoxically speed ventricular conduction over the bypass tract and can trigger VF; the only safe pharmacologic agents slow the ACCESSORY pathway (procainamide, ibutilide).

6. No beta-blocker

WHAT YOU SEE

Crossed-out BETA BUGLE knight

WHAT IT ENCODES

Beta-blockers (metoprolol, esmolol) are AV-nodal blockers and are contraindicated in pre-excited AF — slowing the node shifts conduction to the accessory pathway and increases the risk of VF. Beta-blockers are fine for orthodromic AVRT or ordinary AF, but NOT for the pre-excited, wide-irregular rhythm.

BOARD TRAP

WRONG: 'IV metoprolol will safely slow this fast irregular wide-complex rhythm in a WPW patient.' The discriminator is the accessory pathway — beta-blockade does not slow the bypass tract and removes the node's competition, accelerating ventricular response; procainamide/ibutilide or cardioversion is required instead.

7. No CCB

WHAT YOU SEE

Crossed-out calcium/CCB knight

WHAT IT ENCODES

Non-dihydropyridine calcium-channel blockers (verapamil, diltiazem) are AV-nodal blockers and are contraindicated in pre-excited AF — IV verapamil in particular is a classic board cause of precipitating VF in WPW. They also drop blood pressure, which can worsen hemodynamics during an already fast rhythm.

BOARD TRAP

WRONG: 'IV verapamil is a reasonable choice to control the rate in this irregular wide-complex tachycardia.' The discriminator is pre-excitation — verapamil blocks the node, funnels conduction down the fast accessory pathway, and can precipitate VF and cardiac arrest; it is one of the most-tested wrong answers in WPW-AF.

8. No digoxin

WHAT YOU SEE

Crossed-out COLD-YUM / digoxin knight

WHAT IT ENCODES

Digoxin is contraindicated in pre-excited AF: in addition to its AV-nodal blocking (vagotonic) effect, it can SHORTEN the accessory-pathway refractory period, allowing even faster antegrade conduction and a higher risk of VF. It is therefore doubly dangerous in WPW with AF.

BOARD TRAP

WRONG: 'Digoxin is a gentle, safe option to control ventricular rate in a WPW patient with AF and a low blood pressure.' The discriminator is its effect on the pathway — digoxin both blocks the node AND shortens accessory-pathway refractoriness, accelerating conduction over the bypass tract; it is contraindicated, not a safe alternative.

9. No IV amiodarone

WHAT YOU SEE

Crossed-out IV AMIODARONE

WHAT IT ENCODES

IV amiodarone is best AVOIDED in acute pre-excited AF: it has AV-nodal-blocking properties and reported cases of accelerating the ventricular rate and precipitating VF/hemodynamic collapse, so it is no longer recommended as a safe go-to here despite its general antiarrhythmic versatility. Preferred agents act on the accessory pathway (procainamide, ibutilide), or cardiovert.

BOARD TRAP

WRONG: 'Amiodarone is safe in essentially all wide-complex tachycardias, so it is the best drug for pre-excited AF.' The discriminator is that amiodarone's nodal-blocking effect can paradoxically speed conduction over the pathway in pre-excited AF; the safer choices are procainamide or ibutilide, or synchronized cardioversion.

10. USE procainamide

WHAT YOU SEE

Green-check PROCAINAMIDE knight

WHAT IT ENCODES

Procainamide (class IA) is the preferred drug for hemodynamically STABLE pre-excited AF because it slows conduction in and prolongs the refractory period of the ACCESSORY pathway (and can organize/terminate AF), without preferentially favoring bypass-tract conduction. Dose ~10–17 mg/kg IV at ≤50 mg/min until the arrhythmia terminates, hypotension occurs, or the QRS widens by >50%. It is the safe pharmacologic counterpart to all the crossed-out nodal blockers.

BOARD TRAP

WRONG: 'There is no safe drug for pre-excited AF, so the only option is immediate electrical cardioversion in every case.' The discriminator is hemodynamic stability — an UNSTABLE patient needs cardioversion, but a STABLE patient can be treated pharmacologically with procainamide (or ibutilide), which targets the accessory pathway rather than the node.

11. USE ibutilide

WHAT YOU SEE

Green-check IBUTILIDE knight

WHAT IT ENCODES

Ibutilide (class III, IKr blocker) is an acceptable alternative for stable pre-excited AF — it prolongs accessory-pathway and atrial refractoriness and can chemically cardiovert AF. Dose 1 mg IV over 10 min (0.01 mg/kg if <60 kg), may repeat once. It prolongs the QT, so monitor on telemetry for torsades and correct K⁺/Mg²⁺ before use.

BOARD TRAP

WRONG: 'Ibutilide can be pushed quickly without monitoring because it acts only on the accessory pathway.' The discriminator is QT-driven proarrhythmia — ibutilide prolongs the QT and can cause torsades de pointes, so it must be given over 10 minutes with telemetry and electrolyte repletion, not as a rapid unmonitored bolus.

12. VF → sudden death

WHAT YOU SEE

Skull flag VF→SUDDEN DEATH, 25% of APs

WHAT IT ENCODES

The reason pre-excited AF is feared is that very rapid antegrade conduction over the accessory pathway can degenerate into ventricular fibrillation and cause sudden cardiac death — this is the lethal endpoint that the entire 'no nodal blockers' rule exists to prevent. High-risk pathways (short antegrade refractory period, shortest pre-excited RR <250 ms) are most prone.

BOARD TRAP

WRONG: 'Pre-excited AF is uncomfortable but self-limited and rarely life-threatening, so urgent treatment is unnecessary.' The discriminator is the VF risk — pre-excited AF is a true emergency that can progress to VF/SCD, so it demands prompt correct therapy (procainamide/ibutilide or cardioversion) and definitive ablation.

13. Unstable → cardiovert

WHAT YOU SEE

DEFIB READY box, ALWAYS AVAILABLE

WHAT IT ENCODES

If the patient is hemodynamically UNSTABLE (hypotension, ischemia, shock, altered mental status) — or if there is any diagnostic doubt — perform immediate SYNCHRONIZED cardioversion (e.g., 100–200 J biphasic), with a defibrillator ready in case it deteriorates to VF. Electrical therapy is universally safe in pre-excited AF and is the fallback when drugs fail or are contraindicated.

BOARD TRAP

WRONG: 'Even in an unstable WPW patient with pre-excited AF, try IV antiarrhythmics first before cardioverting.' The discriminator is instability — an unstable patient gets immediate synchronized cardioversion; delaying electrical therapy for a drug trial in a crashing patient is the dangerous wrong answer.

14. Ablation cure

WHAT YOU SEE

ABLATION / VALSALVA / adenosine ladder note

WHAT IT ENCODES

Definitive long-term therapy for symptomatic WPW (and after any episode of pre-excited AF) is catheter ABLATION of the accessory pathway, which is curative with high success and low complication rates and eliminates the SCD risk. (Note: vagal maneuvers/adenosine on this ladder belong to ORTHODROMIC AVRT, the narrow-complex node-dependent rhythm — NOT to pre-excited AF.)

BOARD TRAP

WRONG: 'A patient who survived an episode of pre-excited AF can be managed indefinitely with antiarrhythmic drugs and avoidance of nodal blockers, without ablation.' The discriminator is the demonstrated high-risk, malignant pathway — a manifest pathway that supported pre-excited AF should be ablated, as ablation is curative and removes the ongoing VF/SCD risk that drugs only partially mitigate.

15. Bystander AP → VF

WHAT YOU SEE

Bystander knight, 'AV BLOCK ↑ AP → VF' sign on the right tower

WHAT IT ENCODES

This panel encodes the core mechanism: blocking the AV node creates a one-way street so AF impulses pour down the 'bystander' accessory pathway unimpeded, increasing pathway conduction and driving the ventricular rate toward VF. It visually ties every crossed-out nodal blocker to the lethal endpoint — AV block (drug-induced) → more accessory-pathway conduction → VF.

BOARD TRAP

WRONG: 'Because the accessory pathway is only a passive bystander during AF, slowing the AV node simply controls the rate without harm.' The discriminator is that the bystander pathway is the FAST route — block the node and conduction shifts to it, accelerating the ventricular rate to VF; this is exactly why nodal blockers are contraindicated.